

C# BACKEND TRACK | OOP PHASE

Task 1

# System Analysis Task

---

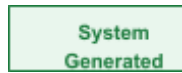
Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

---

## SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:



**System Generated**

The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as `List.Count + 1`.



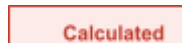
**User Input**

The user types this value directly at the keyboard when prompted by the system.



**Default Value**

A fixed value set by the code when the object is created. The user never chooses it. Example: `isBooked = false`, `status = "Active"`.



**Calculated**

Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.



**From List**

The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.



Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently active loans, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

### Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

| Entity Name: <b>Book</b> |           |                  |
|--------------------------|-----------|------------------|
| Property Name            | Data Type | Source Tag       |
| BookID                   | int       | System Generated |
| BookTitle                | string    | User Input       |
| BookAuthor               | string    | User Input       |
| BookTotalCopies          | string    | User Input       |
| BookAvailableCopies      | int       | Default Value    |
|                          |           |                  |
|                          |           |                  |

| Entity Name: <b>Member</b> |           |                  |
|----------------------------|-----------|------------------|
| Property Name              | Data Type | Source Tag       |
| MemberID                   | int       | System Generated |
| MemberName                 | string    | User Input       |
| MemberEmail                | string    | User Input       |
| MemberPhone                | string    | User Input       |
| MemberRegistrationDate     | string    | System Generated |
| IsMemberActive             | bool      | Default Value    |
|                            |           |                  |

Entity Name: **Loan**

| Property Name      | Data Type | Source Tag        |
|--------------------|-----------|-------------------|
| LoanID             | Int       | System Generated  |
| BookId             | int       | From List         |
| MemberID           | int       | From List         |
| BorrowDate         | string    | System Generated  |
| BookCopyReturnDate | string    | System Calculated |
| ReturnDate         | string    | System Generated  |
| FineAmount         | Double    | System Calculated |
| Status             | string    | Default Value     |

Entity Name: **Reservation**

| Property Name   | Data Type | Source Tag       |
|-----------------|-----------|------------------|
| reservationId   |           | System Generated |
| BookID          |           | From List        |
| MemberID        |           | From List        |
| reservationDate |           | System Generated |
| Status          |           | Default Value    |
|                 |           |                  |
|                 |           |                  |

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

## Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

A Member is registered first and given a unique memberId. Each Book is added with a title, author, genre, and a set number of copies — availableCopies starts equal to totalCopies.

When a Member borrows a Book, the system checks availableCopies > 0. If available, a new Loan is created, the borrowDate is recorded, the dueDate is automatically set to borrowDate + 14 days

When the Member returns the book, status changes to "Returned", and availableCopies is incremented back. If returnDate > dueDate, fineAmount is calculated (number of overdue days × fine rate) and attached to the Loan.

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

**Your task:** For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

### Guest

| Property Name    | Data Type | Source Tag (fill in) |
|------------------|-----------|----------------------|
| guestId          | int       | System Generated     |
| guestName        | string    | User Input           |
| guestPhone       | string    | User Input           |
| guestEmail       | string    | User Input           |
| guestNationality | string    | User Input           |
| registrationDate | string    | System Generated     |

### Room

| Property Name | Data Type | Source Tag (fill in) |
|---------------|-----------|----------------------|
| roomId        | int       | System Generated     |
| roomNumber    | string    | User Input           |

|              |         |                   |
|--------------|---------|-------------------|
| roomType     | string  | User Input        |
| floorNumber  | int     | System Generated  |
| nightlyPrice | decimal | System Calculated |
| isAvailable  | bool    | Default Value     |

| Booking        |           |                      |
|----------------|-----------|----------------------|
| Property Name  | Data Type | Source Tag (fill in) |
| bookingId      | int       | System Generated     |
| guestId        | int       | From List            |
| roomId         | int       | From List            |
| checkInDate    | string    | User Input           |
| checkOutDate   | string    | User Input           |
| numberOfNights | int       | System Calculated    |
| totalPrice     | decimal   | System Calculated    |
| status         | string    | Default Value        |

| CheckInRecord |           |                      |
|---------------|-----------|----------------------|
| Property Name | Data Type | Source Tag (fill in) |
| recordId      | int       | System Generated     |
| bookingId     | int       | From List            |
| guestId       | int       | From List            |
| roomId        | int       | From List            |
| actualCheckIn | string    | System Generated     |

|                             |                      |                      |
|-----------------------------|----------------------|----------------------|
| <code>actualCheckOut</code> | <code>string</code>  | System<br>Generated  |
| <code>finalAmount</code>    | <code>decimal</code> | System<br>Calculated |



### Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

| #  | Service Name    | What it does                                                                                   |
|----|-----------------|------------------------------------------------------------------------------------------------|
| 1  | RegisterMember  | Creates a new Member record and generated ID and today's registration date                     |
| 2  | AddBook         | Adds a new Book and sets availableCopies equal to totalCopies                                  |
| 3  | BorrowBook      | Creates a Loan for a member and book, decrements availableCopies, and sets the 14-day due date |
| 4  | ReturnBook      | Marks a Loan as Returned, restores availableCopies.                                            |
| 5  | CalculateFine   | CalculateFine the fine                                                                         |
| 6  | ReserveBook     | Creates a Reservation for a member                                                             |
| 7  | ViewActiveLoans | Returns all Loan records where status is Active                                                |
| 8  |                 |                                                                                                |
| 9  |                 |                                                                                                |
| 10 |                 |                                                                                                |

### Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

A Guest is registered first, receiving a unique guestId.

When a Guest books a Room, the system verifies isAvailable == true. A new Booking is created linking guestId A Booking can be cancelled before check-in: its status changes to "Cancelled" and the Room's isAvailable .

When the guest checks out, finalAmount is calculated from the actual stay length, and the Room's isAvailable is restored to true so it can be booked again.